

54. Arithmetic Unit for Trigonometric Functions (TFUv1)

This MCU has an arithmetic unit for trigonometric functions (TFU).

The TFU handles the high-speed calculation of $\sin$ f, $\cos$ f, atan2 f, and hypot f functions.

An option must be specified to make the C or C++ compiler produce instructions that use the TFU. For details, refer to the manual for the compiler you are using.

The ICLK is used as the operating clock for the TFU.

54.1 Overview

Table 54.1 lists the specifications of the TFU.

Table 54.1 TFU specifications

Item	Description			
Arithmetic Processing	Calculation of sine, cosine, arctangent, and hypotenuse - A sine and cosine can be simultaneously calculated. - An arctangent and hypotenuse can be simultaneously calculated.			
Range and Unit of Values	Arithmetic Processing	I/O		Range
	Calculating sine	Input	Angle θ	$-\text{float_max} \leq \theta \leq \text{float_max}^{*1}$
		Output	$\sin\theta$	$-1.0 \leq \sin\theta \leq 1.0$
	Calculating cosine	Input	Angle θ	$-\text{float_max} \leq \theta \leq \text{float_max}^{*1}$
		Output	$\cos\theta$	$-1.0 \leq \cos\theta \leq 1.0$
	Calculating arctangent	Input	x and y coordinates	$-\text{float_max} \leq x \leq \text{float_max}^{*1}$ $-\text{float_max} \leq y \leq \text{float_max}^{*1}$
		Output	$\text{atan}(y/x)$	$-\pi \leq \text{atan}(y/x) \leq \pi$
	Calculating hypotenuse	Input	x and y coordinates	$-\text{float_max} \leq x \leq \text{float_max}^{*1}$ $-\text{float_max} \leq y \leq \text{float_max}^{*1}$
		Output	$\sqrt{x^2 + y^2}$	$0 \leq \sqrt{x^2 + y^2} \leq \infty$
Data Type for Processing	Single-precision floating-point			
Number of cycles for calculation	Sine: 14 Cosine: 14 Arctangent: 14 Hypotenuse: 14 + 2 for software processing			

Note 1. float_max is the maximum value that can be expressed as single-precision floating-point: $(2 - 2^{-23}) \times 2^{127}$.